Haleh Damirchi

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Machine learning researcher and Ph.D. graduate from Queen's University with seven years of experience developing deep learning models in Python and PyTorch, including four years in probabilistic and time-series forecasting.

Work Experience

Graduate Research Fellow — May 2021 – April 2026
Queen's University — *AIIM Lab, RCV Lab*

- Co-developed a multitask counting and localization network using pretrained VGG and point-level supervision.
- Enhanced efficiency and accuracy of FPV Pedestrian path prediction by designing a transformer-based context-aware multimodal trajectory forecasting model.
- Improved the accuracy of pedestrian path prediction by designing a socially informed CVAE-based forecaster using online pseudo-trajectory augmentation for challenging training samples.
- Boosted the rigor of model comparisons by conducting an analysis of evaluation methods for trajectory forecasting.

University Lecturer — Jan 2026 – April 2026
Queen's University — *Introduction to Data Science*

- Taught SQL and relational databases, along with data analysis and visualization using Pandas and Matplotlib.
- Covered machine learning topics including dimensionality reduction, imputation, noise reduction, k-fold cross-validation, logistic regression, and classification.

Technical Skills

Programming & Tools: Python, C, SQL, MATLAB, Git, LaTeX, MySQL, FastAPI

Machine Learning: PyTorch, TensorFlow, Keras, scikit-learn

Architectures and Methods: CNNs, RNNs, Transformers, probabilistic forecasting, multimodal learning

Selected Publications

[1] **H. Damirchi**, M. Greenspan and A. Etemad, "A Comprehensive and Critical Analysis of Metrics and Evaluation Practices in Pedestrian Trajectory Forecasting", Under Review.

[2] **H. Damirchi**, A. Etemad and M. Greenspan, "Socially-informed Reconstruction for Pedestrian Trajectory Forecasting", IEEE/CVF Winter Conference on Applications of Computer Vision (WACV 2025).

[3] **H. Damirchi**, M. Greenspan and A. Etemad, "Context-Aware Pedestrian Trajectory Prediction with Multimodal Transformer", IEEE International Conference on Image Processing (ICIP 2023).

[4] M. Zand, **H. Damirchi**, A. Farley and M. Molahasani, "Multiscale Crowd Counting and Localization by Multitask Point Supervision ", IEEE International Conference on Acoustics (ICASSP 2022).

Education

Queen's University, Ph.D. in Electrical and Computer Engineering — 2026
Amirkabir University of Technology, M.Sc. in Electrical Engineering — 2020
University of Tabriz, B.Sc. in Electrical Engineering — 2016

Selected Projects

SlouchLess | *Python*

- Packaged and published standalone Windows and Linux releases for an application that detects prolonged slouching from webcam video and provides real-time posture reminders.
- Built a data collection and training pipeline for learning posture patterns.

PaperScraper | *Python*

- Built a retrieval-augmented LLM pipeline to extract datasets, sample counts, and evaluation metrics from research papers, with citations to the supporting source text.
- Implemented PDF parsing and OCR, HyDE retrieval with embeddings, and schema-constrained LLM outputs for structured information extraction.
- Developed an automated pipeline to evaluate extracted outputs against ground truth using statistical similarity metrics and LLM-as-a-judge scoring.

Speaker Extraction | *TensorFlow, Pytorch, Matlab*

- Enhanced target speaker extraction performance by developing stacked BLSTM models that outperformed DNN and MSE baselines across PESQ, STOI, and SDR.
- Improved speech quality and intelligibility by designing and implementing psychoacoustic loss functions using Mel and Gammatone filter banks and perceptual hearing models.